

Question_6:

Code:

```
1  /*
2  Question No: 6
3  Employees form a hierarchy. Employee 1 is the general director.
4  For every query (x,k), find the boss of employee x that is
5  k levels higher in the hierarchy.
6  If such a boss does not exist, print -1.
7  Binary lifting is used to jump several levels upward quickly.
8  */
9
10 #include <iostream>
11 using namespace std;
12
13 const int N=200005;
14 int up[N][20];
15
16 void link(int i,int j) {
17     up[i][0]=j;
18 }
19
20 int main() {
21     int n,q;
22     cin>>n>>q;
23
24     for(int i=2;i<=n;i++) {
25         int p;
26         cin>>p;
27         link(i,p);
28     }
29     for(int j=1;j<20;j++)
30         for(int i=1;i<=n;i++)
31             up[i][j]=up[up[i][j-1]][j-1];
32
33     while(q--) {
34         int x,k;
35         cin>>x>>k;
36
37         for(int j=19;j>=0;j--) {
38             if(k&(1<<j)) {
```

```
39         x=up[x][j];
40
41         if(x==0) break;
42     }
43 }
44 cout<<(x?x:-1)<<"\n";
45 }
46 }
```

Output:

Test Case_1:

```
5 3
1 3 2 2
4 2
1
1 2
-1
1 3
-1
PS C:\Users\Prabin Yadav\Do
```

Test Case_2:

```
5 3
1 1 3 3
4 1
3
4 2
1
4 3
-1
PS C:\Users\Prabin Yadav\Documen
```